

PERSONAL
INFORMATION

Name: Jiyuan Zhang
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EDUCATION

University of Illinois Urbana-Champaign Urbana, IL
 Ph.D. in Computer Science Aug 2024 – (May 2029)
 Advisor: Prof. [Tianyin Xu](#) and [Nam Sung Kim](#) (co-advising)
University of Illinois Urbana-Champaign Urbana, IL
 M.S. in Computer Science Aug 2022 – May 2024
 Advisor: [Prof. Tianyin Xu](#)
 Thesis: A Software Approach to Accelerating Memory Translation for Virtualized Clouds
New Jersey Institute of Technology Newark, NJ
 B.S. in Computer Science Jan 2020 – May 2022
 GPA: 4.0/4.0

REFEREED
CONFERENCE
PUBLICATIONS

1. **[MICRO '26]** Yang Zhou, Houxiang Ji, Bikrant Sharma, **Jiyuan Zhang**, Yu Li, Linjie Ma, Jaeyong Lee, Myoungjun Chun, Jihong Kim, Sudarsun Kannan, and Nam Sung Kim. “CXL AnySSD: A Composable CXL SSD Using a CXL Type-2 Device and Any SSDs”. In *Proceedings of the 59th IEEE/ACM International Symposium on Microarchitecture (MICRO)*, Oct 2026.
2. **[OSDI '26]** Jongyul Kim, Jaehwan Lee, Inhoe Koo, Peizhe Liu, **Jiyuan Zhang**, Junho Ahn, Tianyin Xu, and Youngjin Kwon. “Oxbow: A Coordinated Architecture for Multi-component File Systems”. In *Proceedings of the 20th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, Jul 2026. [\[Link\]](#)
3. **[OSDI '25]** Siyuan Chai*, **Jiyuan Zhang***, Jongyul Kim, Alan Wang, Fan Chung, Jovan Stojkovic, Weiwei Jia, Dimitrios Skarlatos, Josep Torrellas, and Tianyin Xu. “EMT: An OS Framework for New Memory Translation Architectures”. In *Proceedings of the 19th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, Jul 2025. [\[Link\]](#)
4. **[HotOS '25]** **Jiyuan Zhang**, Jongyul Kim, Chloe Alverti, Peizhe Liu, Weiwei Jia, and Tianyin Xu. “Rethinking Tiered Storage: Talk to File Systems, Not Device Drivers”. In *Proceedings of the ACM SIGOPS 20th Workshop on Hot Topics in Operating Systems (HotOS)*, May 2025. [\[Link\]](#)
5. **[ASPLOS '25]** Yan Sun, Jongyul Kim, Zeduo Yu, **Jiyuan Zhang**, Siyuan Chai, Michael Jaemin Kim, Hwayong Nam, Jaehyun Park, Eojin Na, Yifan Yuan, Ren Wang, Jung Ho Ahn, Tianyin Xu, and Nam Sung Kim. “M5: Mastering Page Migration and Memory Management for CXL-based Tiered Memory Systems”. In *Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, Apr 2025. [\[Link\]](#)
6. **[ASPLOS '24]** **Jiyuan Zhang**, Weiwei Jia, Siyuan Chai, Peizhe Liu, Jongyul Kim, and Tianyin Xu. “Direct Memory Translation for Virtualized Clouds”. In *Proceedings of the 29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, Apr 2024. [\[Link\]](#)
7. **[PACT '23]** Weiwei Jia*, **Jiyuan Zhang***, Jianchen Shan, Yiming Du, Xiaoning Ding and Tianyin Xu. “HugeGPT: Storing Guest Page Tables on Host Huge Pages to Accelerate Address Translation”. In *Proceedings of the 32nd International Conference on Parallel Architectures and Compilation Techniques (PACT)*, Oct 2023. [\[Link\]](#)

8. **[EuroSys '23]** Weiwei Jia*, **Jiyuan Zhang***, Jianchen Shan, and Xiaoning Ding. “Making Dynamic Page Coalescing Effective on Virtualized Clouds”. In *Proceedings of the 18th European Conference on Computer Systems (EuroSys)*, May 2023. [\[Link\]](#)
9. **[ICSE '23]** Wenbo Wang, Tien N. Nguyen, Shaohua Wang, Yi Li, **Jiyuan Zhang**, and Aashish Yadavally. “DeepVD: Toward Class-Separation Features for Neural Network”. In *Proceedings of the 45th ACM/IEEE International Conference on Software Engineering (ICSE)*, May 2023. [\[Link\]](#)
10. **[SoCC '22]** Weiwei Jia, **Jiyuan Zhang**, Jianchen Shan, Jing Li, and Xiaoning Ding. “Achieving Low Latency in Public Edges by Hiding Workloads Mutual Interference”. In *Proceedings of the 13th Symposium on Cloud Computing (SoCC)*, Nov 2022. [\[Link\]](#)

* Co-lead author

REFEREED JOURNAL PUBLICATIONS

1. **[TC '24]** Weiwei Jia*, **Jiyuan Zhang***, Jianchen Shan, and Xiaoning Ding. “Effective Huge Page Strategies for TLB Miss Reduction in Nested Virtualization”. In *IEEE Transactions on Computers (TC)*, 2024. [\[Link\]](#)

* Co-lead author

RESEARCH EXPERIENCE

Networking and Distributed Systems Labs, HPE Labs May 2025 – Present
GPU-Friendly NIC Interface With CXL

- Developing a high-performance, parallelism-friendly NIC interface and supporting hardware for direct GPU-NIC interoperability.

Accelerating NVMe Storage Virtualization Using CXL

- Developing a novel CXL-based virtualization scheme for NVMe devices that supports multi-tenancy, delivers passthrough-level performance, and maintains compatibility with commodity hardware and unmodified VMs.

UIUC xLab, Prof. Tianyin Xu

Aug 2022 – Present

UIUC FAST, Prof. Nam Sung Kim

Jan 2026 – Present

Accelerating Memory Page Allocation for Non-Coherent CXL Shared Memory Pools

- Developing a novel hardware-software architecture that enables high-performance, page-granular, and application-transparent dynamic memory allocation from CXL shared memory pools without hardware cache coherence.

Rethinking Tiered Storage: Talk to File Systems, Not Device Drivers

- Proposed a new file-system architecture that enables storage tiering by directly multiplexing device-specific file systems.

EMT: An OS Framework for New Memory Translation Architectures

- Designed the Extensible Memory Translation (EMT) framework and interface to support new memory translation architectures without sacrificing performance.
- Identified and generalized correctness and performance issues in hash-table-based page-table designs, such as ECPT, and proposed solutions and mitigations.

Direct Memory Translation for Virtualized Clouds

- Developed a novel address translation scheme that reduces worst-case page-table walks to 1, 2, and 3 steps for native, virtualized, and nested-virtualized execution, respectively, while maintaining backward compatibility with modern architectures, including x86, Arm, and RISC-V.
- Proposed direct mapping, which leverages clusters of application Virtual Memory Areas and the OS’s ability to relocate page-table pages to establish a mathematical mapping from each virtual page to its corresponding last-level page-table entry in physical memory.

	<i>HugeGPT: Storing Guest Page Tables on Host Huge Pages to Accelerate Address Translation</i> - Developed a software solution that improves Page-Walk Cache efficiency by strategically clustering page-table pages in physical memory. - Demonstrated that placing guest page tables within host huge pages reduces the number of page-walk steps and improves the Page-Walk Cache hit rate.	
	NJIT Operating System Group, Prof. Xiaoning Ding	Sep 2021 – Aug 2022
	<i>Making Dynamic Page Coalescing Effective on Virtualized Clouds</i> - Identified host-guest page-size mismatch as a major cause of increased TLB misses and performance degradation in virtualized systems. - Developed a software-only solution that mitigates the issue in virtualized systems.	
	<i>Achieving Low Latency in Public Edges by Hiding Workloads Mutual Interference</i> Developed a task scheduler that identifies workload critical paths and performs adaptive scheduling.	
	NJIT SPACE Lab, Prof. Shaohua Wang	May 2021 – Sep 2021
	<i>Identifying Software Vulnerabilities with Graph-Based Neural Networks</i> Designed and implemented an automated toolchain to identify security patches in software sources and extract source-code features for vulnerability classification.	
PROFESSIONAL EXPERIENCE	Hewlett Packard Enterprise	Milpitas, CA
	Research Associate Intern (part- & full-time, HPE Labs)	May 2025 – (Dec 2026)
	University of Illinois Urbana-Champaign	Champaign, IL
	Graduate Research/Teaching Assistant	Aug 2022 – May 2026
	New Jersey Institute of Technology	Newark, NJ
	Undergraduate Research Assistant	Jan 2022 – May 2022
AWARDS AND HONORS	Wing Kai Cheng Fellowship, UIUC	2024 - 2025
	NJIT President’s Medal for Academic Excellence, NJIT	2022
	Summa Cum Laude, NJIT	2022
	Dean’s List, NJIT	2020 - 2022
TALKS AND PRESENTATIONS	EMT: An OS Framework for New Memory Translation Architectures	
	USENIX Symposium on Operating Systems Design and Implementation [Conf. talk] (Boston, USA), Jul 9, 2025	
	Rethinking Tiered Storage: Talk to File Systems, Not Device Drivers	
	The ACM SIGOPS Workshop on Hot Topics in Operating Systems [Conf. talk] (Banff, Canada), May 14, 2025	
	Direct Memory Translation for Virtualized Clouds	
	- Cornell University [Invited] (Ithaca, New York), Oct 10, 2024 - ACM Int’l. Conf. on Architectural Support for Programming Languages and Operating Systems [Conf. talk] (San Diego, USA), May 1, 2024	
	HugeGPT: Storing Guest Page Tables on Host Huge Pages to Accelerate Address Translation	
	Int’l. Conf. on Parallel Architectures and Compilation Techniques [Conf. talk] (Vienna, Austria), Oct 23, 2023	
TEACHING AND MENTORING EXPERIENCE	Research Mentoring	
	Peizhe Liu (Graduate Student, UIUC) I mentored Liu on the project of Direct Memory Translation for Virtualized Clouds and Rethinking Tiered Storage: Talk to File Systems, Not Device Drivers.	Oct 2023 – Aug 2025

- Siyuan Chai (Graduate Student, UIUC) Aug 2022 – Apr 2025
I mentored Chai on the project of EMT: An OS Framework for New Memory Translation Architectures.
- Fan Chung (Undergraduate Student, UIUC) Jan 2023 – Aug 2024
I mentored Chung on the project of EMT: An OS Framework for New Memory Translation Architectures.
- Yiming Du (Graduate Student, University of Rhode Island) Aug 2022 – May 2023
I mentored Du on the project of HugeGPT: Storing Guest Page Tables on Host Huge Pages to Accelerate Address Translation.

Teaching Assistant

- UIUC CS 423: Operating Systems Design Fall 2023
Worked with Prof. Tianyin Xu
- NJIT CS 114: Introduction to Computer Science II Spring 2021
Worked with Prof. Calvin M. James

GRANTS Travel grants for OSDI '25, HotOS '25, ASPLOS '24, OSDI '23, and EuroSys '23

SERVICES Program Committee: MICRO '26
Artifact Evaluation Committee: SOSP '23

REFERENCES

Tianyin Xu

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Associate Professor, Department of Computer Science
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